Applications of linear systems

1 $a = \$1.45$

2 $m = 0.9 \text{ km}$

3
a $w = 1.6$
b 5 m

4
a $x + y = 172$
b $x - y = 18$
c $x = 95$
d $y = 77$

5
a $y = 3x - 260\,000, y = x + 480\,000$
b $x = \$370\,000, y = \$850\,000$

6
a $2f + 3g = 18.30$
b $g = \dfrac{44.03 - 5f}{7}$
c $f = \$3.99$
d $g = \$3.44$

7
a $y = 10x$
b $y = 7x + 18$
c $x = 6, y = 60$

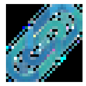
8 $x = 40, y = 8$

9 $x = 63, y = 7$

10
a $x + y = 16\,000$
b $0.14x - 0.06y = 700$
c $x = \$8300, y = \$7700$

11
a $64x - 3y = 372$
b $8x - 15y = -12$
c $x = 6, y = 4$
d 396 units
e 108 units

12
a $9x + 2y = 37$
b $3x + 8y = 49$
c $x = 3, y = 5$
d 17 units
e 31 units

13 a $25x + 4y = 70$ b $5x + 12y = 70$  $x = 2, y = 5$ d 30 units

e 50 units

14

a $x = 2y - 15$ b $2x + 4y = 50$ c $x = 5, y = 10$

15

a $5x + 3y = 25$ b $8x + 7y = 82$

c $x = -\frac{71}{11}, y = \frac{210}{11}$

16

a $c = 5.90 - xk$ b $k = \$0.50$ c $c = \$1.90$

17

a $x = 8$ b 2017

18

a $x = 2y$ b $x + y = 90$ c $x = 60^\circ, y = 30^\circ$

19

a $R + C = 9$ b $6.45R + 5.75C = 55.25$

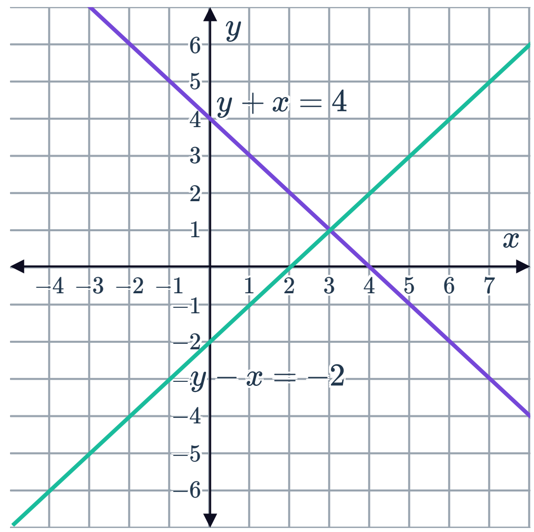
c $R = 5, C = 4$

Additional questions

20

a Equation 1: $y + x = 4$
Equation 2: $y - x = -2$

b

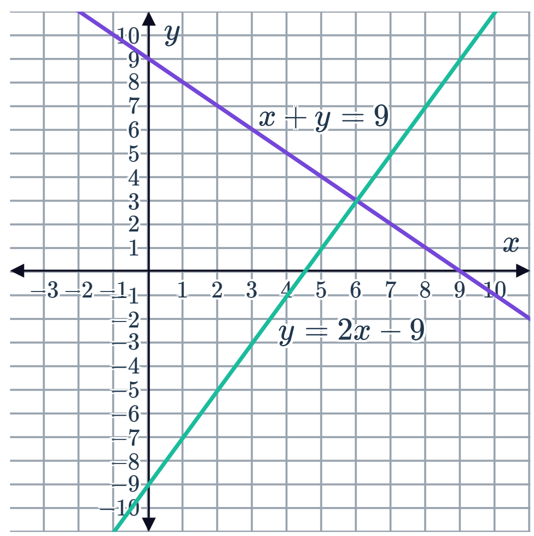


c $x = 3, y = 1$

21

a Equation 1: $x + y = 9$
Equation 2: $y = 2x - 9$

b

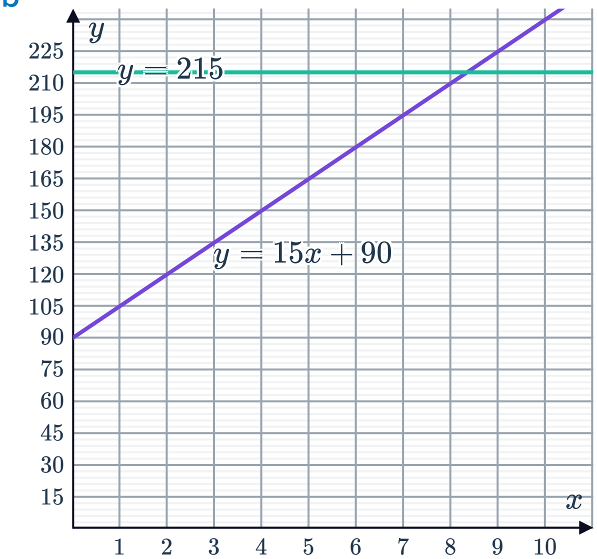


c Lauren did 3 parts of the assignment and Nicole did 6 parts.

22

a $y = 15x + 90,\ y = 215$

b



c No

23

a Any context where positive fractions are valid solutions. For example, the weights of items (such as fruit).

b Any answer where negative integers are valid for both the independent and dependent variable. For example, a situation where x is the number of days before or after a specific date, and y is the outside temperature in degrees fahrenheight.

c Any answer where the independent variable can be a negative integer and the dependent variable can be a positive integer. For example the daily balance in two different bank accounts based on different savings and spending habits.

24

a The solution to the system $(4.8, 4.2)$ is not feasible because you cannot have part of a hippo or a rhinocerous.

b The solution results in 4.8 rhinos and 4.2 hippopotami. Depending on the circumstances, you might be able to round to 5 rhinos and 4 hippopotami in order to make sense in context.

25

a $9x + 5y = 57, 3x + 10y = 69$

b Each chocolate croissants costs $\$3$ and each guava roll costs $\$6$.

26

a $9f + 7g = 30.51,\ 5f + 8g = 24.72$

b Oysters cost $f = \$1.92$ each and prawns cost $g = \$1.89$ each.

27

a 190 student tickets and 160 general admission tickets

b Answers will vary.
By choosing the elimination method, we can multiply the equation $x+y=350$ by 3 and combine it with the equation $3x+5.5y=1450$ by means of subtraction. This is an efficient method to solve for y and once y is known, it can be substituted into either original equation to solve for x.

28

a Oxtail stew costs $\$16.25$ and sweet potato pie costs $\$5.99$.

b Answers will vary.
By choosing the substitution method, we can rewrite the first equation as $y=38.49-2x$ by 3 and substitute this into the second equation, $5x+3y=99.22$. This is an efficient method to solve for x and once x is known, it can be substituted into either original equation to solve for y.

29 Answers will vary. Sample answers provided.

a (iii) would make this solution nonviable, since we cannot have a rectangle with a length (or width) of 0 ft.

b (ii) would make this solution nonviable, since we cannot have a decimal number of cats and mice.

c (iv) would make this solution nonviable, since we cannot have a negative quantity of birds.

d (i) would make this solution nonviable - although we can have a negative temperature, we cannot have a negative number of dust particles.